ROOM 3: INTRO TO LOG ANALYSIS

Learning Objective

The objective of this room was to understand different log analysis methods and learn how to extract useful information from logs during a security investigation.

1. Introduction

In this room, I learned how log analysis helps identify important events from large amounts of log data. I explored different methods and tools used to search logs, extract information, and identify suspicious activities.

2. Key Concepts Learned

Manual Log Analysis

Manual analysis involves opening log files and searching for specific entries using commands or text editors. It is useful when an analyst needs to inspect particular events or investigate a small set of logs.

Automated Log Analysis

Automated analysis uses tools or scripts to process log files and return useful information. It can save time when analysing large amounts of data.

Command-Line Log Analysis

Linux commands such as `grep`, `cut`, `sort`, `uniq`, and `wc` help filter, extract, sort, and count information from log files.

Regular Expressions (Regex)

Regex patterns help find information that matches a specific format, such as IP addresses, URLs, or particular values inside log entries.

Log Parsing and Normalisation

Parsing extracts useful fields from raw log data. Normalisation converts the extracted information into a consistent format so it can be searched and analysed more easily.

Sigma and YARA

Sigma uses YAML-based rules to describe log-based detection logic. YARA is used to identify patterns in files and other data, especially during malware analysis.

3. Practical Learning

During this room, I worked with log files and practised extracting information using command-line tools and regular expressions.

I learned how to identify IP addresses, search for specific requests, count HTTP response codes, and decode Base64-encoded values using CyberChef.

I also explored Sigma rules and YARA rules and learned how they help security professionals identify suspicious patterns.

4. SOC Analyst Perspective

Log analysis helps SOC analysts investigate alerts and identify unusual behaviour. For example, an analyst can examine web server logs to identify repeated requests, suspicious URLs, or unexpected source IP addresses.

Filtering and counting log entries can help an analyst understand the scope of an activity and decide whether further investigation is required.

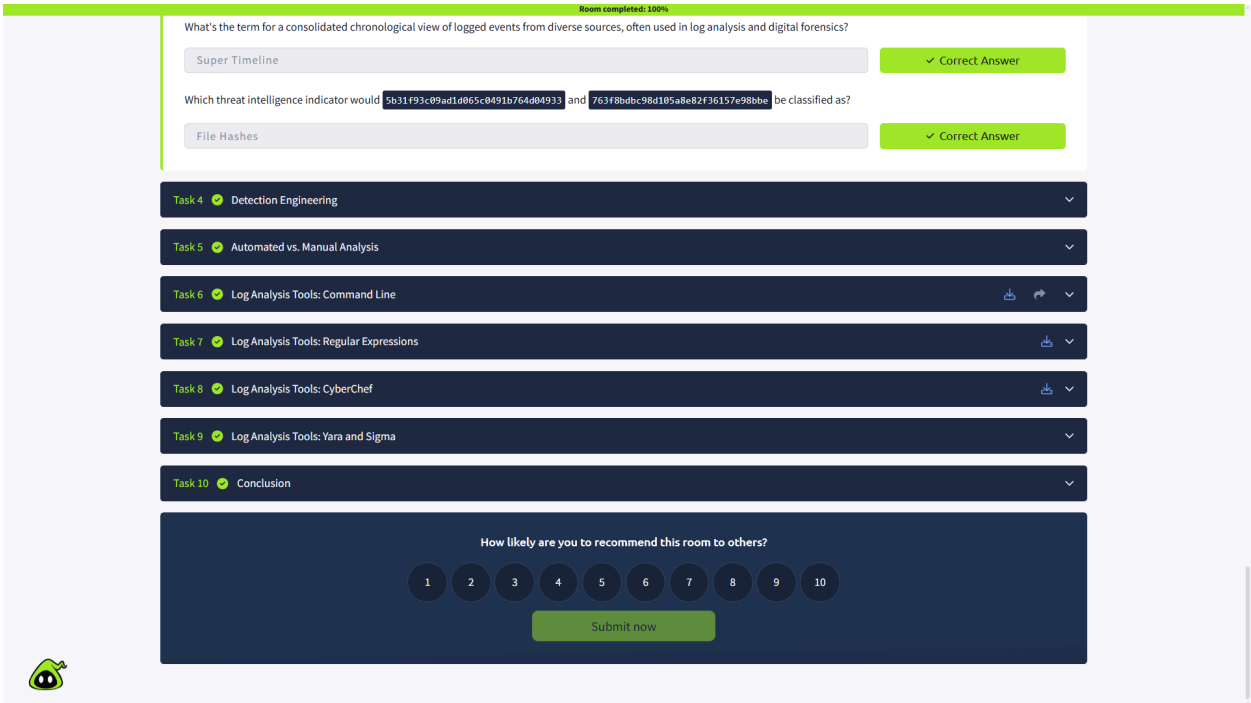
5. What I Learned

- I understood the difference between manual and automated log analysis.
- I learned how to use Linux commands to filter and extract log information.
- I practised using regex to identify specific patterns.
- I learned how to extract useful information from web server logs.
- I understood the basic purpose of Sigma and YARA rules.
- I learned how log analysis supports security investigations.

6. Conclusion

This room helped me improve my understanding of log analysis and the tools used to investigate log data. I learned that filtering, searching, and extracting relevant information are important skills for a SOC analyst.

7. Evidence



Room completed: 100%

Answer the questions below

What is the default file path to view logs regarding HTTP requests on an Nginx server?

/var/log/nginx/access.log ✓ Correct Answer

A log entry containing `%2E%2E%2F%2E%2E%2Fproc%2Fself%2Fenviron` was identified. What kind of attack might this infer?

Path Traversal ✓ Correct Answer

Task 5 Automated vs. Manual Analysis

Task 6 Log Analysis Tools: Command Line

Task 7 Log Analysis Tools: Regular Expressions

Task 8 Log Analysis Tools: CyberChef

Task 9 Log Analysis Tools: Yara and Sigma

Task 10 Conclusion

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Allows for contextual analysis. The analyst has a broader understanding of the organization and cyber security landscape.	N/A

Answer the questions below

A log file is processed by a tool which returns an output. What form of analysis is this?

Automated ✓ Correct Answer

An analyst opens a log file and searches for events. What form of analysis is this?

Manual ✓ Correct Answer

Task 6 Log Analysis Tools: Command Line

Task 7 Log Analysis Tools: Regular Expressions

Task 8 Log Analysis Tools: CyberChef

Task 9 Log Analysis Tools: Yara and Sigma

Task 10 Conclusion

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solution, like the Elastic (ELK) Stack or Splunk, can be more efficient and offer additional log analysis and visualization features. However, the command line remains essential for quick and straightforward log analysis tasks.

Answer the questions below

Use `cut` on the `apache.log` file to return only the URLs. What is the flag that is returned in one of the unique entries?

c701d43cc5a3acb9b5b04db7f1be94f6 — ✓ Correct Answer

In the `apache.log` file, how many total HTTP 200 responses were logged?

52 — ✓ Correct Answer

In the `apache.log` file, which IP address generated the most traffic?

145.76.33.201 — ✓ Correct Answer

What is the complete timestamp of the entry where `110.122.65.76` accessed `/login.php`?

31/Jul/2023:12:34:40 +0000 — ✓ Correct Answer

Task 7 Log Analysis Tools: Regular Expressions

Task 8 Log Analysis Tools: CyberChef

Task 9 Log Analysis Tools: Yara and Sigma

Task 10 Conclusion

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```
}
```

In the configuration above, we use our previously defined regular expression pattern to extract IPv4 addresses from the "message" field of incoming log events. The extracted values will be added under the custom "ipv4_addresses" field name we defined. Typically, IP addresses are extracted automatically by default configurations. But this simple example shows the power of regular expression patterns when dealing with complex log files and custom field requirements.

The Logstash room and the official Grok documentation are fantastic resources for further exploring Logstash input and filter configurations!

Answer the questions below

How would you modify the original `grep` pattern above to match blog posts with an ID between 20-29?

post=2[0-9] — ✓ Correct Answer

What is the name of the filter plugin used in Logstash to parse unstructured log data?

Grok — ✓ Correct Answer

Task 8 Log Analysis Tools: CyberChef

Task 9 Log Analysis Tools: Yara and Sigma

Task 10 Conclusion

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Answer the questions below

Locate the "loganalysis.zip" file under `/root/Rooms/introloganalysis/task8` and extract the contents.

No answer needed — ✓ Correct Answer

Upload the log file named "access.log" to CyberChef. Use regex to list all of the IP addresses. What is the full IP address beginning in 212?

212.14.17.145 — ✓ Correct Answer

Using the same log file from Question #2, a request was made that is encoded in base64. What is the decoded value?

THM{CYBERCHEF_WIZARD} — ✓ Correct Answer

Using CyberChef, decode the file named "encodedflag.txt" and use regex to extract by MAC address. What is the extracted value?

08-2E-9A-4B-7F-61 — ✓ Correct Answer

Task 9 ✓ Log Analysis Tools: Yara and Sigma

Task 10 ✓ Conclusion

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This YARA rule can be expanded to look for:

- Multiple IP addresses
- IP Addresses based on a range (for example, an ASN or a subnet)
- IP addresses in HEX
- If an IP address lists more than a certain amount (I.e., alert if an IP address is found five times)
- And combined with other rules. For example, if an IP address visits a specific page or does a certain action

If you want to learn more about Yara, check out the Yara room on TryHackMe.

Answer the questions below

What languages does Sigma use?

YAML — ✓ Correct Answer

What keyword is used to denote the "title" of a Sigma rule?

title — ✓ Correct Answer

What keyword is used to denote the "name" of a rule in YARA?

rule — ✓ Correct Answer

Task 10 ✓ Conclusion

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